5.3.24

AI25BTECH11030 -Sarvesh Tamgade

Question:Solve the following pair of linear equations:

$$3x - 5y = 4 \tag{0.1}$$

$$2y + 7 = 9x \tag{0.2}$$

Solution:
The equation of a line:

$$\mathbf{n^T x} = c \tag{0.3}$$

Line L:

$$\begin{pmatrix} 3 & -5 \end{pmatrix} \mathbf{x} = 4 \tag{0.4}$$

Line K:

$$\begin{pmatrix} 9 & -2 \end{pmatrix} \mathbf{x} = 7 \tag{0.5}$$

Combining both the equations:

$$\begin{pmatrix} 3 & -5 \\ 9 & -2 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 4 \\ 7 \end{pmatrix} \tag{0.6}$$

Writing the augmented matrix:

$$\left(\begin{array}{cc|c} 3 & -5 & 4 \\ 9 & -2 & 7 \end{array}\right) \tag{0.7}$$

Perform the row operation $R_2 \to R_2 - 3R_1$:

$$\left(\begin{array}{cc|c} 3 & -5 & 4 \\ 0 & 13 & -5 \end{array}\right) \tag{0.8}$$

Perform the row operation $R_2 \to \frac{1}{13} R_2$:

$$\left(\begin{array}{cc|c} 3 & -5 & 4 \\ 0 & 1 & -\frac{5}{13} \end{array}\right) \tag{0.9}$$

Back substitute using $R_1 \to R_1 + 5R_2$:

$$\left(\begin{array}{cc|c} 3 & 0 & \frac{27}{13} \\ 0 & 1 & -\frac{5}{13} \end{array}\right) \tag{0.10}$$

Therefore, the solution vector is:

$$\mathbf{x} = \begin{pmatrix} \frac{9}{13} \\ -\frac{5}{13} \end{pmatrix} \tag{0.11}$$

Final Answer: Therefore, the solution vector is:

$$\boxed{\mathbf{x} = \begin{pmatrix} \frac{9}{13} \\ -\frac{5}{13} \end{pmatrix}}$$

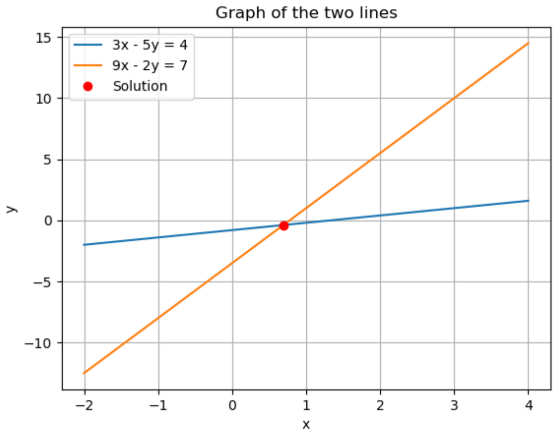

Fig. 0.1: Vector Representation